apsara

Date: __/__/____

SUB QUERIES

1. What is a sub query?

A subquery is a query written inside another SQL query.

- Inner Query : The query inside another query
- Outer Query : The main query that uses the inner query's result.

→ First, SQL finds an answer to a smaller questio
→ Then it uses that answer to solve the main question

Syntax:

```
SELECT column_name
FROM table_name
WHERE column_name = (
    SELECT column_name
    FROM table_name
    WHERE condition
);
```

The inner query is placed inside parentheses ()

2. **When do we use subqueries?**

We use subqueries when we need to fetch

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data frome one table based on information obtained from a query on another table or from the same table.

Ex: find the employee earning the highest salary

Suppose we have an employee table.

emp_name	salary
Ravi	20,000
Priya	30,000
Arun	45,000
Meena	25,000

```
SELECT emp_name, salary
FROM employee
WHERE salary = (
    SELECT MAX (salary)
    FROM employee
);
```

3. How does a subquery execute?

Step 1: Inner query

Find the required value

Ex: Highest salary = 50000

↓

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Step 2 : outer query
use the value to retrieve matching row
Find the employee earning 50,000

4. Types of subqueries

(i) Single row subquery
(ii) Multi row subquery
(iii) Correlated subquery

(i) Single row subquery

A subquery that returns only one row (one result row)
use the (=) operator when the inner query returns a single value

Ex: Employee table

emp_name	department	salary
Ravi	IT	30,000
Priya	HR	25,000
Arun	IT	45,000
Meena	HR	35,000

Find the employee who earns the highest salary

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```
SELECT emp_name , salary
FROM employee
WHERE salary = (
    SELECT MAX ( salary)
    FROM employee
);
```

Inner query : MAX (salary) returns 45000
Outer query : finds the employee whose salary equals 45000

o/p :

emp_name	salary
Arun	45,000

(ii) Multi row subquery

A subquery that returns more than one row
else the (IN) operator to match a value against multiple results.

find employees who work in departments that have employees earning more than ₹40,000

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```
SELECT emp_name , department , salary
FROM employees
WHERE department IN (
    SELECT department
    FROM employees
    WHERE salary > 40000
);
```

→ Inner query finds the departments of employees earning more than ₹40,000
→ It returns IT, because Arun earns ₹45,000
→ Outer query returns all employees whose department is IT

o/p:

emp_name	department	salary
Ravi	IT	30,000
Arun	IT	45,000

(iii) Correlated subquery

A subquery that refers to a column or value from the outer query unlike a simple subquery, it depends on the current row being considered by the outer query

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Ex: find employees earning more than their department's average salary.

```
SELECT emp_name, department, salary
FROM employee e1
WHERE salary > (
    SELECT AVG (salary)
    FROM employee e2
    WHERE e1.department = e2.department
);
```

Department	salaries	Average
IT	30,000 ; 45,000	37,500
HR	25,000 ; 35,000	30,000

Now compare each employee's salary with their department average.

o/p:

emp_name	department	salary
Arun	IT	45,000
Meena	HR	35,000

→ Arun earns more than the IT average and
→ Meena earns more than the HR average

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Type	Inner query returns	common operator or feature
* Single-row	one row	=
* Multi-row	multiple rows	IN
* Correlated	Depends on the outer row	References outer query

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